The final products of digestion of carbohydrates are glucose, fructose and galactose, these products are absorbed in the intestines. The non – digestible carbohydrates present in the food such as cellulose, hemicelluloses, pentosans, galactans, fructosans etc add bulk to the contents of large intestine and are excreted in the faeces.

Storage, metabolism of carbohydrates

Glucose, galactose and fructose absorbed in the intestines pass through the portal circulation to the liver. In the liver a part of the glucose and the entire galactose and fructose are converted into glycogen. A portion of glucose enters into the general circulation and to the various tissues for being oxidized and used as energy. A small portion of the glucose is stored in liver and muscle as glycogen and some portion of the glucose is converted into fat and stored in adipose tissue. The oxidation of glucose in the tissues occurs in two stages as indicated below;

1. Glycogen $\longrightarrow$ Glucose $\longrightarrow$ Pyruvic acid $\longrightarrow$ Lactic acid

2. Pyruvic acid $\xrightarrow{\text{Oxidation}}$ $CO_2 + H_2O$

The first stage is called 'glycolysis. The oxidation of pyruvic acid takes place through a series of reactions known as tricarboxylic acid cycle (Krebs's cycle).



Drinking tea / coffee can severely inhibit your body's ability to absorbs vitamins and minerals – especially Iron.

Deficiencies

A deficiency of carbohydrates makes the body to utilize fats for energy, if it is not rectified it leads to ketone bodies formation which occurs due to oxidation of fats.

Excessive carbohydrates

Excessive consumption of carbohydrates leads to heart disease, diabetes, and obesity.

5.6 Fat

The name fat may make it sound like something you shouldn't eat. But fat is an important part of a healthy diet. Fat from your diet gives you energy. As a bonus, fat in food helps you feel full, so you don't eat too much.

Some foods, including most fruits and vegetables, have almost no fat. Other foods have plenty of fat. They include nuts, oils, butter, and meats like beef.

The lipids are a heterogeneous group of substances found in plant and animal tissues, which share the property of being relatively insoluble in water, and soluble in organic solvents, such as ether, chloroform and benzene. Fat contain 9 kcal per gram.

Functions of fats

1. They are the chief energy stores of the body, which form an important source of energy during starvation or other emergencies.
2. Fats play a role in the absorption of fats soluble vitamins like vitamins A, D, E and K.
3. Fats are the constituents of cell membrane structure and regulate the membrane permeability.
4. Subcutaneous fat acts as an insulator and helps in retaining body heat.
5. Fats are important as cellular metabolic regulators (Steroid hormones and prostaglandin).
6. Fat is the main energy store in the body and the most concentrated source of energy in the diet – 1 g of fat provides 37kJ (9Kcal), more than double that provided by either protein or carbohydrate (4Kcal).

Unit_5_Applied_nutrition_Gen_Nur.indd 114 04-03-2020 17:12:18